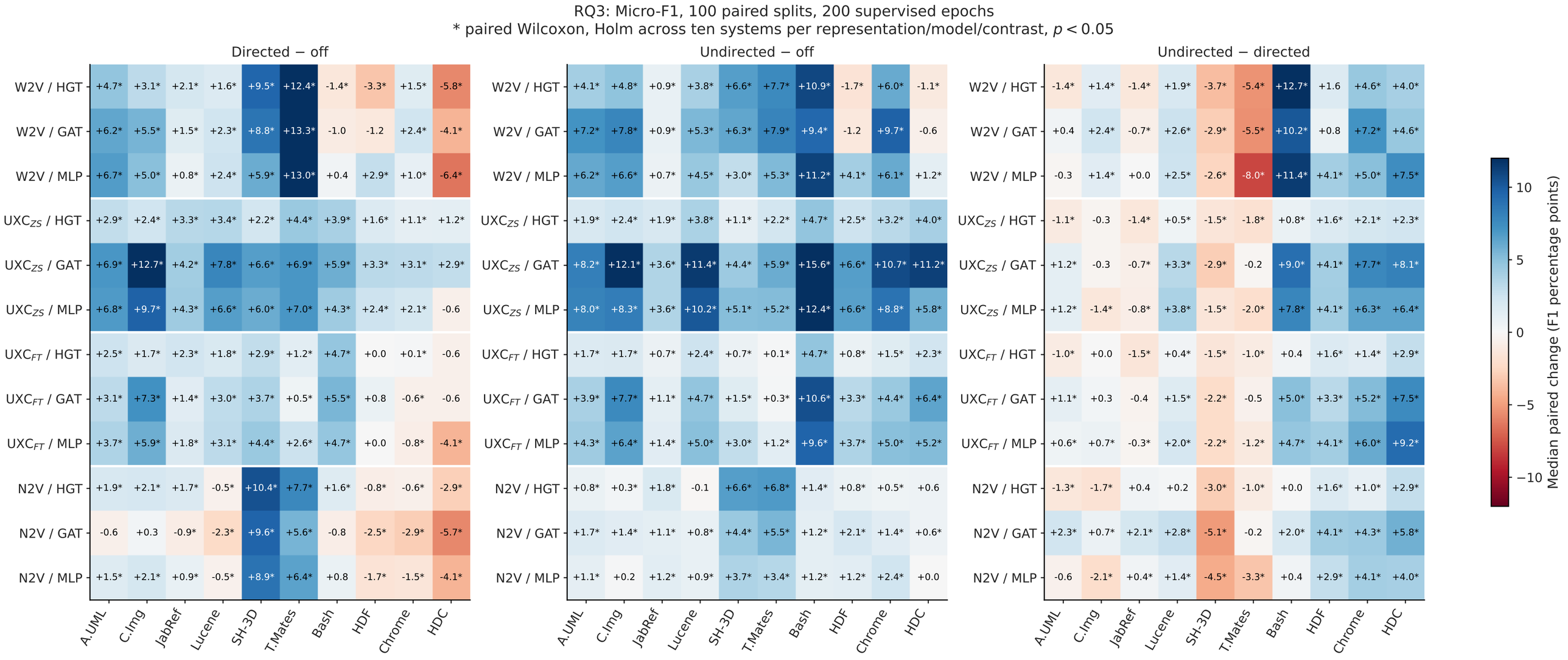
RQ3: Micro-F1, 100 paired splits, 200 supervised epochs
* paired Wilcoxon, Holm across ten systems per representation/model/contrast, $p < 0.05$
Directed − off
Undirected − off
Undirected − directed
W2V / HGT +4.7* +3.1* +2.1* +1.6* +9.5* +12.4* -1.4* -3.3* +1.5* -5.8*
W2V / GAT +6.2* +5.5* +1.5* +2.3* +8.8* +13.3* -1.0 -1.2 +2.4* -4.1*
W2V / MLP +6.7* +5.0* +0.8* +2.4* +5.9* +13.0* +0.4 +2.9* +1.0* -6.4*
UXC$_{ZS}$ / HGT +2.9* +2.4* +3.3* +3.4* +2.2* +4.4* +3.9* +1.6* +1.1* +1.2*
UXC$_{ZS}$ / GAT +6.9* +12.7* +4.2* +7.8* +6.6* +6.9* +5.9* +3.3* +3.1* +2.9*
UXC$_{ZS}$ / MLP +6.8* +9.7* +4.3* +6.6* +6.0* +7.0* +4.3* +2.4* +2.1* -0.6
UXC$_{FT}$ / HGT +2.5* +1.7* +2.3* +1.8* +2.9* +1.2* +4.7* +0.0 +0.1* -0.6
UXC$_{FT}$ / GAT +3.1* +7.3* +1.4* +3.0* +3.7* +0.5* +5.5* +0.8 -0.6* -0.6
UXC$_{FT}$ / MLP +3.7* +5.9* +1.8* +3.1* +4.4* +2.6* +4.7* +0.0 -0.8* -4.1*
N2V / HGT +1.9* +2.1* +1.7* -0.5* +10.4* +7.7* +1.6* -0.8* -0.6* -2.9*
N2V / GAT -0.6 +0.3 -0.9* -2.3* +9.6* +5.6* -0.8 -2.5* -2.9* -5.7*
N2V / MLP +1.5* +2.1* +0.9* -0.5* +8.9* +6.4* +0.8 -1.7* -1.5* -4.1*
W2V / HGT +4.1* +4.8* +0.9* +3.8* +6.6* +7.7* +10.9* -1.7* +6.0* -1.1*
W2V / GAT +7.2* +7.8* +0.9* +5.3* +6.3* +7.9* +9.4* -1.2 +9.7* -0.6
W2V / MLP +6.2* +6.6* +0.7* +4.5* +3.0* +5.3* +11.2* +4.1* +6.1* +1.2*
UXC$_{ZS}$ / HGT +1.9* +2.4* +1.9* +3.8* +1.1* +2.2* +4.7* +2.5* +3.2* +4.0*
UXC$_{ZS}$ / GAT +8.2* +12.1* +3.6* +11.4* +4.4* +5.9* +15.6* +6.6* +10.7* +11.2*
UXC$_{ZS}$ / MLP +8.0* +8.3* +3.6* +10.2* +5.1* +5.2* +12.4* +6.6* +8.8* +5.8*
UXC$_{FT}$ / HGT +1.7* +1.7* +0.7* +2.4* +0.7* +0.1* +4.7* +0.8* +1.5* +2.3*
UXC$_{FT}$ / GAT +3.9* +7.7* +1.1* +4.7* +1.5* +0.3* +10.6* +3.3* +4.4* +6.4*
UXC$_{FT}$ / MLP +4.3* +6.4* +1.4* +5.0* +3.0* +1.2* +9.6* +3.7* +5.0* +5.2*
N2V / HGT +0.8* +0.3* +1.8* -0.1 +6.6* +6.8* +1.4* +0.8* +0.5* +0.6
N2V / GAT +1.7* +1.4* +1.1* +0.8* +4.4* +5.5* +1.2* +2.1* +1.4* +0.6*
N2V / MLP +1.1* +0.2 +1.2* +0.9* +3.7* +3.4* +1.2* +1.2* +2.4* +0.0
W2V / HGT -1.4* +1.4* -1.4* +1.9* -3.7* -5.4* +12.7* +1.6 +4.6* +4.0*
W2V / GAT +0.4 +2.4* -0.7* +2.6* -2.9* -5.5* +10.2* +0.8 +7.2* +4.6*
W2V / MLP -0.3 +1.4* +0.0 +2.5* -2.6* -8.0* +11.4* +4.1* +5.0* +7.5*
UXC$_{ZS}$ / HGT -1.1* -0.3 -1.4* +0.5* -1.5* -1.8* +0.8* +1.6* +2.1* +2.3*
UXC$_{ZS}$ / GAT +1.2* -0.3 -0.7* +3.3* -2.9* -0.2 +9.0* +4.1* +7.7* +8.1*
UXC$_{ZS}$ / MLP +1.2* -1.4* -0.8* +3.8* -1.5* -2.0* +7.8* +4.1* +6.3* +6.4*
UXC$_{FT}$ / HGT -1.0* +0.0 -1.5* +0.4* -1.5* -1.0* +0.4 +1.6* +1.4* +2.9*
UXC$_{FT}$ / GAT +1.1* +0.3 -0.4 +1.5* -2.2* -0.5 +5.0* +3.3* +5.2* +7.5*
UXC$_{FT}$ / MLP +0.6* +0.7* -0.3* +2.0* -2.2* -1.2* +4.7* +4.1* +6.0* +9.2*
N2V / HGT -1.3* -1.7* +0.4 +0.2 -3.0* -1.0* +0.0 +1.6* +1.0* +2.9*
N2V / GAT +2.3* +0.7* +2.1* +2.8* -5.1* -0.2 +2.0* +4.1* +4.3* +5.8*
N2V / MLP -0.6 -2.1* +0.4* +1.4* -4.5* -3.3* +0.4 +2.9* +4.1* +4.0*
A.UML C.Img JabRef Lucene SH-3D T.Mates Bash HDF Chrome HDC
Median paired change (F1 percentage points)
10
5
0
−5
−10